



Community, Children, Child: New Directions for Equitable Delivery Of High-Quality Early Childhood Education and Care (ECEC)

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Abstract

Theory and measurement approaches to Early Childhood Education and Care (ECEC) quality focus disproportionately on educator inputs. Yet, ECEC quality is profoundly relational. Each child and the collective of children in the ECEC classroom contribute to the nature of educators' daily work and the resources they require to enable high-quality learning experiences. We review the literature examining the contribution of the child, and collective of children in a class on ECEC quality and identify design and measurement strategies to advance understanding of the ways in which high-quality ECEC can be co-created. We discuss implications for proportionate universal policy approaches to support educator wellbeing and attendant practice to deliver on the promise of ECEC to redress inequities.

Keywords Early childhood education and care · Interactional quality · Classroom composition · Child contributions · Educator-child interactions

Introduction

Children are not passive recipients of environment but actively contribute to shaping their experiences and interactions (Plomin, 2016; Plomin & Bergeman, 1991; Rutter, 2006). While research on parenting has long acknowledged that children are not only shaped by parents, but also shape parenting (Karraker & Coleman, 2005; Tao et al., 2024), far fewer studies (e.g., Vitiello et al., 2012; Zigler & Hall, 1994) have considered the *inputs* of the child and collective of children in ECEC services in the early years of life. Yet, when children enter the door of their ECEC service

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they bring with them their unique individual characteristics and experiences of their community and home. These have the potential to impact their behaviours and capacity to engage in learning (Coppens & Rogoff, 2022; Evans et al., 2014; Froelich & Gerstein, 2025), for better or worse (e.g., Coley et al., 2019). As ECEC is a group setting, the characteristics and circumstances of *all* children in the group can influence the demands placed on their educators and their attendant capacity to facilitate high-quality responsive learning experiences (Diebold & Perren, 2026; Miller et al., 2017; Searle et al., 2024; Yudron et al., 2014). When classroom and educator resources are more limited, or even equal in classrooms where children need more individualised supports (e.g. a standard staff ratio), there will inevitably be inequity.

Children spend long hours in ECEC services. These hours matter. The growth of government investment in ECEC and in children's participation is based on the promise that ECEC can deliver optimal developmental opportunities, especially for those who experience circumstances in their home and community environments that may limit their educational prospects (Burchinal et al., 2015; OECD, 2025). In this paper we review literature examining the inputs of the child and the collective of children on ECEC quality in current systems that focus on standard provision. We identify directions in design and measurement to investigate the ways in which each child, the collective of children and their community influence ECEC quality. We examine the implications for policy and practice in delivering on the promise of ECEC to redress social inequity. Importantly, the aim of this paper is not to assert that children, families or communities have deficits, but that ECEC should be responsive to the requirements of each. Responsiveness may require different levels and types of resourcing to achieve equity.

Relationship at the Centre

Teaching and learning centres on the relationship between educator and child. This relationship is critical in the formative early years when children have their first group-based educational experience outside the home. Interactions in the ECEC context set the foundations of ongoing learning, extending beyond acquisition of constrained curriculum content, to include formative unconstrained skills: learning to regulate emotions, learning to get on with others and learning to learn. Such skills have enduring effects (Bailey et al., 2020; Burchinal et al., 2024; Kang et al., 2019; Rankin et al., 2022). For this reason, theoretical approaches to ECEC quality focus on the *interactional* experiences within the ECEC classroom. While structural characteristics such as curriculum, physical resources, number of staff and staff qualifications are enablers of quality, the body of research evidence places interactional quality, between educator and child, as the key predictor of child development (Downer et al., 2010a, b; Hamre, 2014; Nguyen et al., 2020; Pianta et al., 2016). Edwards and Raikes (2002) describe this relationship as “a dance” between educator and child, capturing the profoundly relational nature of ECEC quality. This metaphor aptly describes

ECEC quality as the product of a reciprocal and bidirectional relationship, in which both child and adult have the possibility for leading roles. That is educators, each child, and the collective of children co-create quality.

Educators undoubtedly make highly significant contributions to ECEC quality (Guedes et al., 2020; Otero-Mayer et al., 2025). For this reason, research studies have focused on educator actions and their role in determining the quality of ECEC provision. This approach is seen in standard measurement and in intervention studies. Standard global measures embody standard definitions of ECEC quality and heavily weight educator actions. For example, in the Instructional Support domain in CLASS-PreK (Pianta et al., 2008) there are four dimensions rated. The description of three of these dimensions focus on *educator actions* with each descriptor commencing with the phrase “the teacher (action)...”. Only one dimension (Quality of Feedback), focuses on educator-child interaction, requiring a focus on educator response to the child using the commencing phrase “The educator and child...”. Similarly, ECERS-R (Harms et al., 1998) rates educator actions and provisions. For example, “Staff model good social skills (Ex. Are kind to others, listen, empathise, cooperate), “Staff help with resources to enhance play (Ex. Help set up obstacle course for tricycles)” and “Staff usually take action to cut down on the spread of germs.”. Moreover, standard measures place a strong weighting on verbal interactions (Thorpe et al., 2020). The consequence is twofold. First, preferencing verbal interactions limits capacity to capture educator non-verbal strategies (e.g., pauses and silences) that encourage children to participate in and drive learning (Thorpe et al., 2020). Yet, these strategies are associated with deep learning (Dillon, 2006; Siraj et al., 2016). Second, the focus on verbal inputs may fail to capture cultural variation in relationships norms and interaction style (Ringsmose et al., 2022; Rogoff et al., 1993; Wolf & Avornyo, 2023).

The limitations of standard global measures of quality may explain why intervention studies, even within dominant culture, have had limited impact (Egert et al., 2018; Thorpe et al., 2020). These studies have sought to change educators’ behaviours to align with the definitions of high quality captured in standard global measures. Yet, such interventions while effective in changing educator behaviours, have delivered negligible effect on children’s developmental outcomes (Egert et al., 2018; Pianta et al., 2016). The reason for these disappointing outcomes likely relates to the understanding of quality as *standard*, thus failing to consider the capacity of educators to be responsive to the requirements of each child and the collective of children within the class group. All educators’ face policy and practice constraints (e.g., Harper et al., 2025; Robinson et al., 2021; Thorpe et al., 2024) but in circumstances where demands on the educator are high, these will have greater effect on their capacity (Panthi et al., 2025; Partee et al., 2020). Moreover, each child within a class group may not encounter the same experience. Some children attract more educator attention. Thus, recent studies that have focused on individual children’s experience within the ECEC sector, using standard measures such as InCLASS (Downer et al., 2010a, b), have yielded better prediction of child outcomes than those using global measures (e.g., Slot et al., 2023).

Successful learning occurs in an environment in which an educator is *responsive* to the abilities and behaviours of each individual child and the collective of children within a class group (Hamre, 2014; McNally & Slutsky, 2017; Nguyen et

al., 2020). Limitations to delivery of high-quality ECEC occurs when educators are not sufficiently personally and professionally resourced to support each child and their collective (Bull et al., 2024; Kwon et al., 2022; Panthi et al., 2025). Implicit in standard measures and conceptualisations of ECEC quality is a ‘typical’ educator workday. Yet, the composition of many ECEC class groups changes daily, and the behaviours of educators and children can change hour-to-hour as they move across varied activities that incur different levels of demand (Thorpe et al., 2020; Vitiello et al., 2012).

Context matters. In Fig. 1 we present the key contextual factors hypothesised to determine ECEC quality. In contrast to an ecological model that presents community and family as distal influences (Bronfenbrenner & Morris, 2007), we propose these experiences enter the classroom with the two key players in the educational relationship. The differing social circumstances and learning experiences of child and educator determine *both* child requirements and educator capacity to respond to these.

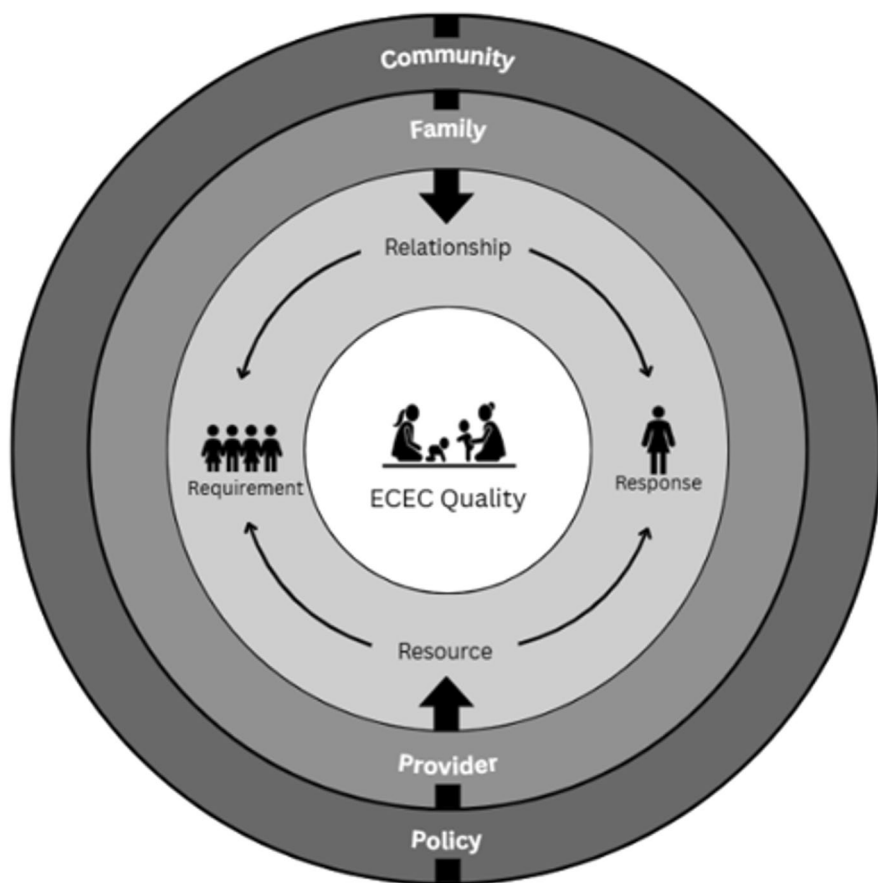


Fig. 1 Classroom ecological model depicting contextual factors determining quality within the ECEC sector (adapted from Thorpe et al., 2021, classroom ecological model)

Class Composition and Demand on Educator

To date, in understanding the association of the class group and interactional quality in ECEC, research studies have taken two broad approaches. First, they have indexed composition through demographic measures comprising child age, ability, socio-economic circumstance, minority language or race status and gender. They have assessed the association of mean value of such characteristics with observational measures of ECEC quality. Second, using these indices some have examined the heterogeneity/homogeneity of the class group by testing the association with dispersal, commonly measured as proportion. To summarise this literature, we expanded a review of the association of class composition in ECEC by Aguiar and Aguiar (2020) applying a digital search replicating their search terms (i.e., (a) “early childhood education and care” OR “center-based child care” OR preschool* OR “3-to-5-year* old*” AND teacher* OR educator* OR professional* AND (b) “class* composition” OR “class* characteristics” OR “class* heterogeneity” OR “group homogeneity” AND (c) “class* observations” OR “observed interaction*” OR “observed practice*” OR “process quality”) and data bases (i.e. EBSCO data bases such as Academic Search Complete, ERIC, PsycARTICLES, PsycINFO, as well as Scopus and Web of Science) to extend coverage to 2020–2025. Our inclusion criteria replicated those of Aguiar and Aguiar’s (2020), specifically, classrooms of children enrolled in ECEC centre-based programs, focused on classroom composition. Any type of study that provided empirical evidence on observed classroom quality and those that used standardized observation practices and measures were included. Finally, the additional text(s) identified that met inclusion criteria were subject to full text analysis. In Table 1 we summarise the combined findings.

Broadly, with exception of gender for which no reported significant associations were found, the studies report a significant association of demand on educator (indexed by younger child age, higher learning needs, greater disadvantage and minority status) with reduced interactional quality. Notably, age and ability had equal numbers of studies reporting significant and null association potentially reflecting resourcing or homogeneity of class groups. The results across studies converge on the conclusion that heterogeneity of class composition in each of these measures, potentially another index of demand on educator, is associated with poorer standard observed interactional quality. We note that across the three decades covered by the combined reviews (1995–2025) the number of papers examining the association of demographic indices of class composition and standard observed interactional quality rose steadily until a peak between 2010 and 2020 (Fig. 2). Beyond that date, there has been a precipitous decline to a single paper published in 2021 (e.g., Yang et al., 2022). We hypothesise that this decline may be recognition that research designs focusing on demographic characteristic have reached the limits of their value. While consistently replicated, the findings are limited in their capacity to inform actions in policy or practice actions that might improve quality and redress inequities.

Extant studies, while describing a problem of inequity, fail to capture the complexity of the mechanisms connecting interactional quality and class composition. We argue this is for three key reasons. First, demographic variables serve only as a distal proxy for class-group demand on educators. They do not measure intersectionality

Table 1 Statistical information of literature in review

Index	No. of papers	Location	Measure of quality	Measure of composition	Finding
Age	5*	USA=2* Other=3	InCLASS=1 CLASS=4 ECERS=1 Other=2	Heterogeneity Age diversity $N=1$ Classroom Mean Age $N=2$ Number of 6-year-olds $N=1$ Proportion of 3-year-olds vs. 4-year-olds $N=1$ Age composition % * $N=1$	Significant association $N=2$ No association $N=3$
Ability	4	USA=4 Other=0	CLASS=1 ECERS=3 Other=1	Heterogeneity Proportion of children with IEP $N=1$ Percentage of children with IEP $N=1$ Number of children with disability $N=1$ Two subgroups of children with IEP's $N=1$	Significant association $N=2$ No association $N=2$
Disadvantage	11	USA=10 Other=1	CLASS=10 ECERS=5 Other=1	Heterogeneity % of children living in poverty by zip code $N=1$ % of low SES children $N=2$ % of children living below the 150% poverty line $N=2$ Average income-to-needs ratio $N=1$ Mean level of maternal education $N=2$ Proportion of children living below the poverty line $N=1$ % of children in poverty below 130% poverty line $N=1$ Mean SES (average between family income and maternal education) $N=1$ % of children eligible for free lunch $N=1$ Poor children living below the 150% poverty line vs. children non-poor $N=1$	Significant association $N=9$ No association $N=2$
Minority Status	17	USA=13 Other=4	CLASS=11 ECERS=5 Other=5	Heterogeneity % of Ethnicity $N=7$ % of Dual Language Learners $N=5$ % of Non-English Speakers $N=1$ % with low English proficiency $N=1$ % with low majority language proficiency $N=1$ Proportion of Ethnicity $N=5$ Bilingual class groups $N=1$ Predominantly one Ethnicity of children $N=4$	Significant association $N=12$ No association $N=5$
Gender	3	USA=2 Other=1	CLASS=1 ECERS=NA Other=1	Heterogeneity % of Females $N=1$ % of Males $N=2$	Significant association $N=0$ No association $N=3$

This table displays the elements of Aguiar and Aguiar's (2020) systematic review findings and our combined literature search findings between 2020–2025. The standard measures of quality include InCLASS; CLASS; ECERS; Other includes ASSIST; ECCOM; ELLCO; CIS; ESCAPE; APEC. *Age composition literature has had Yang et al. (2022) study added to the count

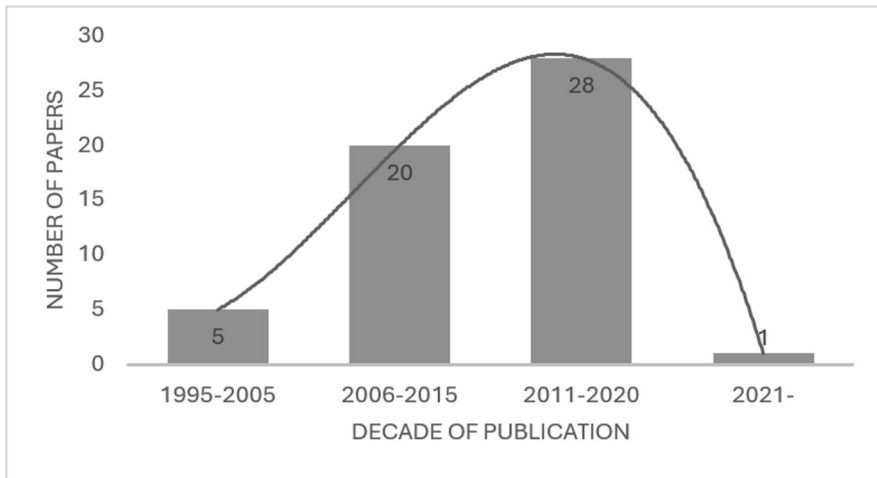


Fig. 2 Research timeline. *Note.* This figure displays the combined literature search findings from Aguiar and Aguiar's (2020) systematic review and the updated findings from our search. Search was conducted on the same EBSCO and Scopus and Web of Science databases

nor directly index child requirements. Second, though a few of the studies include co-variables of classroom resourcing and teacher characteristics, these are generally absent or limited and therefore fail to adequately capture the 'fit' of child requirement and educator capacity to respond (Li et al., 2025). Finally, the generalised demographic composition cannot capture the individual child's influence on an educator or classmates. A single child with high level of demand may disrupt a whole class (e.g., Goulas et al., 2024; Houts et al., 2010; Panthi et al., 2025; Thorpe et al., 2024) and engage disproportionate amount of educator time reducing global quality as assessed by standard measures (Goulas et al., 2024; Howard et al., 2024; Panthi et al., 2025; Thorpe et al., 2024). Importantly, a focus on child behavioural characteristics provides direction for action and contrasts with a focus on demographic composition. While the possibility of changing the demographic composition of a class is likely infeasible, and potentially unethical, educator response to behavioural characteristics of individual children within a class offer a modifiable mechanism connecting class composition, quality of provision and children's development. Missing in the body of studies is the consideration of resources required to respond effectively to the experiences and characteristics of the child, such as reactivity (Belsky & Pluess, 2009; Belsky et al., 2007), response to trauma (Adubasim & Ugwu, 2019), learning difficulties (Jeon et al., 2022), or challenges to behavioural expectation (Panthi et al., 2025) that may explain both individual experiences and group dynamics in the ECEC classroom.

The Child, the Educator and ECEC Quality

"... children require effort, and some children require more effort than others."
Houts et al. (2010)

Each child in a classroom not only contributes to shaping their own social and learning experiences in an ECEC classroom but also the experience of others (Goulas et al., 2024; Miller et al., 2017; Yudron et al., 2014). In studies of ECEC there are many studies that have assessed the association of child behavioural challenges and generalised experiences in the classroom, measured using standard measures of quality. However, these have considered child behaviour as an outcome measure rather than as a predictor of quality (D'Souza et al., 2023) and recent outcomes suggest the need to explore interactions with individual characteristics (Oppermann et al., 2023).

Some studies deriving from the school sector examine the association of child behaviour on classroom learning experiences. Though of older children, these studies are informative in demonstrating mechanism and raising hypotheses of similar effects in the ECEC sector. Goulas et al., (2024) in an experimental study that manipulated class grouping in primary school settings found that those placed in classrooms including children with behavioural difficulties had poorer academic outcomes. Houts and colleagues (2010), using a powerful behavioural-genetic twin design of children aged 5 to 12 years found that children's challenging behaviour (e.g., irritability, negativity, impulsivity, distractibility, and hyperactivity), but not ability (IQ), was the major source of demand on the teacher. In examining the implications of this finding, the authors note that one child presenting behavioural challenges in a classroom can influence the experiences of all other classmates by limiting the time others can access the educator. We note that such behavioural challenges, while potentially genetically driven, are often social in origin, for example predicted by adverse and traumatic events (Adubasim & Ugwu, 2019), hunger (Gallegos et al., 2021; Searle et al., 2024), and tiredness (Hoyniak et al., 2024). Regardless of underlying cause, the important point made is that a child who is displaying disruptive behaviour can redirect an educator's work. An educator's capacity to respond effectively without adequate instructional and emotional resource can be limited (Buettner et al., 2016; Bulotsky-Shearer et al., 2012; Panthi et al., 2025; Staton et al., 2022).

Children who place higher demand on an ECEC classroom can diminish their educator's well-being (Boyd, 2013; Panthi et al., 2025; Zhao & Jeon, 2023; Zheng et al., 2024; Zinsser & Curby, 2014; Zinsser et al., 2016) and may even be a catalyst for staff turnover (Buettner et al., 2016; Klassen & Chiu, 2011; Penttinen et al., 2020). Across the last decade there has been a growing body of literature examining educator wellbeing. An educator's well-being, emotional exhaustion and burnout is associated with poorer quality interactions with children while attendant staff turnover can disrupt attachment relationships (Ansari et al., 2022; Buettner et al., 2016; Hakanen et al., 2006). Recent studies have identified that challenging behaviour of children is among the most salient catalysts for poor educator well-being and staff turnover (Choi et al., 2019; Thorpe et al., 2020). Together, the literature examining aspects of child demand and educator well-being suggests individual child behaviours matter, influencing the experiences of all others in the ECEC classroom.

Advancing knowledge to actionable change requires a focus on the child within the class. Notably, in our review of the literature on class composition, the single paper

focused in ECEC published since 2020 examines the impact of age composition by moving to an individual child measure of behaviour, InCLASS (Yang et al., 2022). The results of this study showed that younger child age was related to lower teacher engagement and that teacher engagement mediated the association between age composition and language gains. The study underscores the imperative to understand the mechanisms connecting interactional quality using direct observational measurement of child behaviours across the complexity of ECEC days. An example from a school setting is provided by Heinschel et al. (2024). In a study of the same teacher working with different math classes, the authors were able to quantify the relative contribution of class composition against teacher instructional quality. They report that student characteristics contributed 7–13% of variance in instructional quality across individual lessons. The limitation of this study was that measurement of instructional quality was via student report, presenting a potential confound. However, the design approach provides a model for examining the potential distorting effects of class composition on instructional quality by capitalising on variation across classes taught by the same teacher.

The example of Heinschel et al. (2024) study raises two hypotheses for research in the context of ECEC. First, unlike secondary schooling where students move between multiple teachers in a single day, in ECEC children typically remain with the same teaching team (lead and assistant educator) throughout a full day. This continuity may amplify the input of child behavioural characteristics on educator wellbeing and capacity to respond effectively. Second, though ECEC does not have defined ‘lessons’, the response of children in different activities of the day may vary with attendant association with educator demand and quality of interactions (Thorpe et al., 2020; Vitiello et al., 2012).

Implications for Research: Innovation in Measurement and Design

The emerging literature directs attention to two key components of method required to identify mechanisms connecting each child, the collective of children on a given ECEC day, classroom interactional quality and child development outcomes. First, understanding individual inputs of a child, and the collective of children to ECEC quality requires finer grained, individualised measurement of each child’s input in the context of their ECEC class group and the varied activities across the ECEC day. Second, understanding the contribution of child characteristics requires research designs that disaggregate child and educator inputs.

Standard measurement of ECEC quality is global. These measures disproportionately weight *educator actions* rather than *responsive interactions*. In studies comparing global measures with those examining individual child experiences within a ECEC group, prediction of child development outcomes is of higher effect size when the individual child experience is the focus (Burchinal et al., 2021; Jeon et al., 2022). The emergence of individualised measures provides the potential to understand individual child experiences in their ECEC group. Currently, such measures use counts (Burchinal et al., 2021) or time sequence selection approaches to sample the child’s

experience (Downer et al., 2010a, b; Slot et al., 2023) thus capturing interactions across different activities. Ongoing research may help to streamline such measures by identifying key events or activities in the ECEC day in which to observe individual child experiences, improving comparability (Thorpe et al., 2020). Individualised child observation measures are thus essential but their development and application an ongoing focus for research. Research designs assessing contributions of the child on ECEC quality and, in turn, developmental outcomes are correlational, applying global measures averaged across groups. Innovation is needed. Drawing on the design strategies of behavioural genetics studies in which, for example, monozygotic twins provide the opportunity to hold genetics constant, we propose design strategies, that hold educator or child ‘constant’ within the ECEC context. There are three designs that could be implemented in this way. First, capitalising on child variability, designs holding educator constant and measuring interactions in different class composition could be applied. In the context of ECEC, particularly daycare services, this would be a **same educator different day design** as composition changes daily. Second, a **same educator different event design** would serve to provide finer scale understanding. In these designs the same educator(s) would be observed interacting with different children and different class groups in specific ECEC activities or events that present different levels of educator demand. There is evidence that some activities present greater general demand on educator (Farran et al., 2017; Thorpe et al., 2020). For example, transitions, mealtimes and sleep times are a source of greater challenge for educators while free play is less demanding (Farran et al., 2017; Thorpe et al., 2020). Times of high demand are associated with reduced observed quality, but also are those with greater variability (Thorpe et al., 2020) and have been identified as barometers both of educator responsiveness and potential demand-reducing strategies (e.g., Pattinson et al., 2014). Children with different levels of reactivity within these events may give insight into why this is so and how improvements can be achieved. Third, taking a different approach, the child can be the constant - a **same child different educator design**. Following a group of focal children longitudinally as they transition to new educators provides opportunity not only to assess the association of child outcomes with changing levels of global quality (e.g., Rankin et al., 2022; Watts et al., 2021), but also to understand the inputs of child behavioural response to educator and classmates (Figs. 3, 4 and 5).

Implications for Policy

The distribution of children requiring a higher level of educator support when attending ECEC is not equal across communities, personal circumstance or levels of disadvantage (Tang et al., 2024; Votruba-Drzal et al., 2004). To take account of the variation in characteristics of communities, children, and child, necessitates moving beyond a quality standard to consider equitable resourcing requirements. Proportionate universalism (Marmot, 2025), in which all children receive access to ECEC services but with greater intensity of resourcing for services experiencing higher levels of complexity and educator demand is a proposed policy mechanism (Gray et al., 2026). There are some examples of targeted interventions that provide

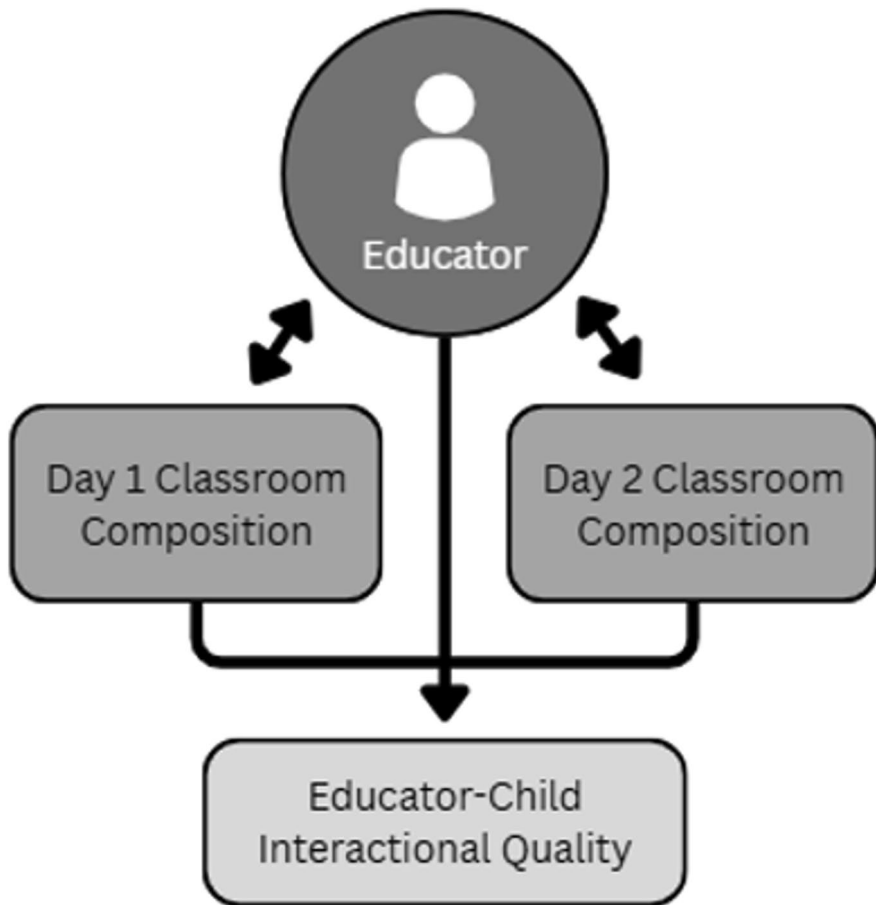


Fig. 3 Visual representation of same educator, different day

intensive supports in ECEC services (e.g. Baulos et al., 2024; Newman et al., 2022) serving highly disadvantaged communities. These demonstrate a capacity to achieve improved equity, seen in child outcomes, through ECEC provisions and family supports and provide strong empirical support for such an approach (e.g., Newman et al., 2022).

The emerging question is how proportionate universalism can be enacted. To do so would require measurement to inform resource distribution and placement of targeted programs. At the broadest level administrative data on socio-economic distribution might be used to facilitate tiered distribution. In some nations, for example Canada and Australia, measurement of child development at school entry provides an index of ‘developmental vulnerability’ for small local areas (AEDC, 2026). These may serve as a more refined guide to tailor community requirements (Janus et al., 2021). However, such methods are focused on geographic areas and may not capture all children and families who require additional supports (Gray et al., 2026). Location of children who require additional support is not uniformly within low socio-

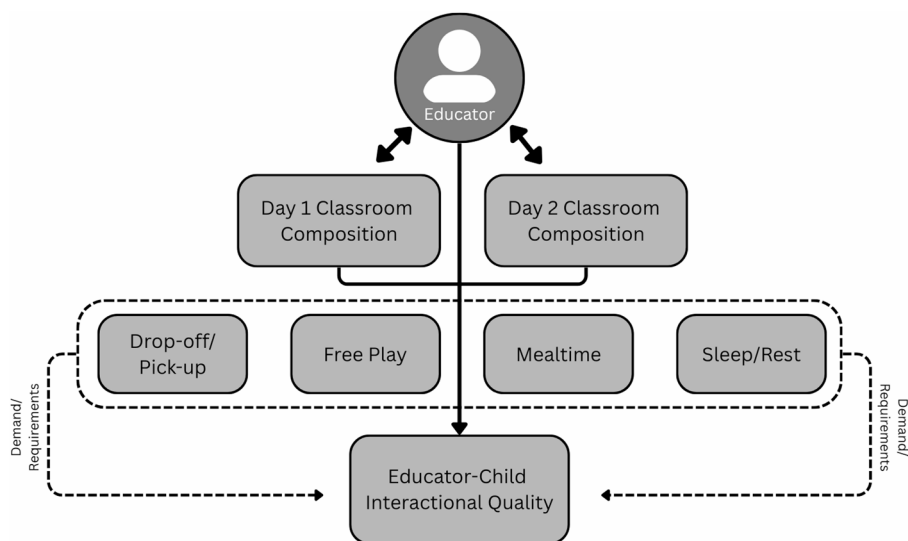


Fig. 4 Visual representation of same educator, different event

economic communities, nor those in which there is a greater average developmental vulnerability (Boddy et al., 2026; Gray et al., 2026; Sammons et al., 2023).

Leveraging of Quality Rating and Improvement Systems (QRIS) provides a finer grained potential to equitable allocation of resources in ECEC. Developed economies not only invest in ECEC services, but in QRIS to ensure these investments are delivering services of sufficient quality to support children's development. Early approaches to QRIS focused on a test: *assessment against a global standard*. Commensurately, public facing ratings of service quality was conceptualised as a method to motivate quality improvement in an educational system that operates as a competitive market (Brogaard & Helby Petersen, 2022; Sabol, 2021; Thorpe et al., 2022). While well-intentioned, this approach has been costly (Staton et al., 2025; Thorpe et al., 2025), inefficient (Australian Government Productivity Commission, 2024; Sabol, 2021; Staton et al., 2025; Thorpe et al., 2025), and subject to manipulation in which some organisations with multiple services 'play the game' by performing quality (Grant et al., 2018; Phillips & Fenech., 2023; Staton et al., 2025; Thorpe et al., 2025); including moving resources and staff across services (Staton et al., 2025). Assessment and rating processes have been reported as stressful to educators and assessors alike (Staton et al., 2025) and a distraction from providing optimal experiences for children (Grant et al., 2018; Thorpe et al., 2025).

Emerging approaches to QRIS are focusing on *quality improvement as a collaboration* with services and their educators (Alston, 2025). These approaches present opportunity for localised information on resource requirements from families, service providers and educators to respond to the dynamics of individual children and the collective of children within a service. Resource allocation can thereby promote help-seeking from services and individual educators to enable co-creation of quality between government agencies and ECEC providers. Trialling and evaluation of such approaches are recommended.

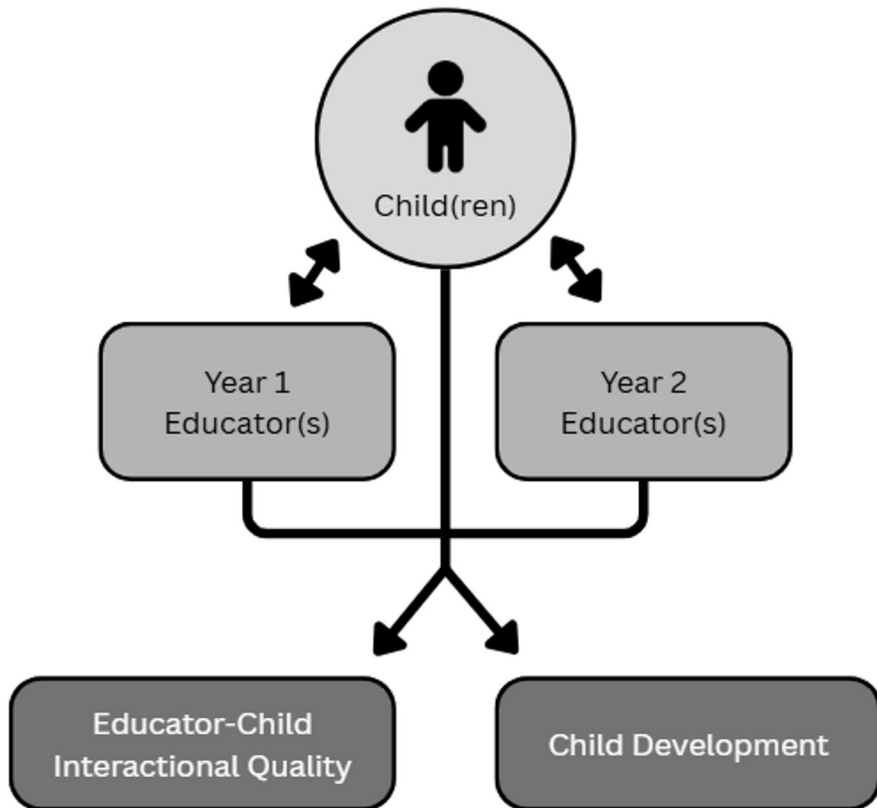


Fig. 5 Visual representation of same child, different educator

Implications for Practice

Research examining quality improvement in practice has disproportionately focussed on standardising educator actions. Yet, interventions focussed on changing educators' behaviours to align with those defined as "high quality" in global standard measures have been largely ineffective (Egert et al., 2018; Pianta et al., 2016). While educator knowledge and skill are undoubtedly important, the work of an ECEC educator is profoundly relational and co-created in interactions with children. Successful learning occurs in an environment in which an educator can be *responsive* to the knowledge, development, and circumstances of each individual child (Hamre, 2014; McNally & Slutsky, 2017; Nguyen et al., 2020). The circumstances, expectations, knowledge and skills the child brings into the classroom have been relatively silenced. Also silenced is the context in which an educator works, including the demands that limit, and the supports that enable, responsive practice. New approaches that consider quality as co-created and therefore consider child contributions to quality not only promise to increase prediction of child outcomes (Burchinal et al., 2021) but direct practice actions to supporting educators (Campbell-Barr & Bogatić, 2017; Hunkin, 2018; Kwon et al., 2022; Thorpe et al., 2021).

There is now a growing body of evidence that informs practice that enables communities, child and children to receive responsive, high-quality ECEC within the classroom. These necessarily focus on *both* educator and child. The well-being of educator's has consistently been reported as associated with their capacity to be responsive to children (Jeon et al., 2018; Kwon et al., 2022) and is influenced by both their personal circumstances (McDonald, Thorpe, & Irvine, 2018) and organisational decisions (Bull et al., 2024; Thorpe et al., 2020; Thorpe et al., 2024). The majority of ECEC educators are women whose pay conditions are generally poor with associations with their health and personal agency reported (McDonald, Thorpe, & Irvine, 2018; Thorpe et al., 2020; Thorpe et al., 2024). Moves to improve the pay, work conditions and emotional supports of educators is undoubtedly central to quality improvement. For those working in more demanding circumstances this is especially so (Thorpe et al., 2024). Organisational decisions about use of their resources, among these leadership (Thorpe et al., 2020; Zhou & Fenech, 2022) staffing levels (Slot, 2018), educator agency (Panthi et al., 2025; Thorpe et al., 2024), and tailored professional development (Warner-Richter et al., 2020) are predictors of job satisfaction and well-being and enablers of responsive interactions that define high-quality ECEC. Alongside, the well-being of children influences their capacity to be agentic learners. For example, children who are food insecure are unable to take advantage of learning opportunities if they arrive at their ECEC service hungry, as hunger limits a child's attention and behavioural regulation capacities (Searle et al., 2024). In this case, resourcing of food provision is an essential component of the ECEC program and should not be at the cost other provisions such as staffing levels (Thorpe et al., 2022). Beyond the ECEC service, connection to multiple support agencies and families are critical in delivery of responsive ECEC programs. Deep engagement with children's families and communities' takes time and resource that is often unrecognised (Dickerson et al., 2025; Tadeu & Lopes, 2025). Acknowledging that standard resourcing is inequitable is central to achieving equitable outcomes for educators, children, and their families.

Conclusion

When children and educators enter the ECEC service they do not leave their personal and community circumstances at the door. Effective ECEC is responsive to each child and the varied circumstances the collective of children brings with them into their ECEC classroom. To deliver on the promise that ECEC offers of promoting children's development and redressing inequity (Burchinal et al., 2022) requires moving beyond *one-size-fits* all models of ECEC quality. High-quality ECEC is fundamentally responsive. Achievement of high-quality interactional experiences, defined as those that are intellectually stimulating, emotionally supportive and predictable, requires not only appropriate training of educators but tailored resourcing provisions to enable educators to enact high quality at the frontline. Research advances are required to elucidate child contributions to quality and inform policy and practice interventions that can deliver equitable educational opportunity.

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